STUDENT 2 · DEFENCE PREPARATION

Your Results, Fully Explained

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

This guide takes Chapter 3 of the thesis one table at a time. Every table is reproduced exactly as it appears in the thesis. For each one you get:

- **the question** the table answers,
- **the test** that was used and **why that test**,
- **how to read every column**,
- **a walk-through of real rows** with the numbers worked out,
- **what it means** in plain words, **what to say**, and **what you may be asked**.

Yellow rows in the tables are the rows you must know.

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1. Part A · Six ideas every result uses
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Part A · Six ideas every result uses

Every number in Chapter 3 is one of these six things. Learn them once and every table becomes readable.

1. Two kinds of variable decide everything

Categorical	Groups or yes/no: gender, uses AI (yes/no), elevated health anxiety (yes/no). Described with **n and %**.
Continuous (a score)	A number on a scale: SHAI-14 (0–42), CSS-12 (12–60). Described with **mean ± SD** or **median (IQR)**.

Which test you use depends only on **what kind of variable you are comparing with what** (Part B).

2. The p-value: "could this be luck?"

Imagine there is really no difference or no link. The p-value is the chance of getting a result at least as strong as ours just by luck. If $p < 0.05$ we call it **significant**: unlikely to be luck.

EXAMPLE

Health anxiety and cyberchondria: r = 0.47, **p < 0.001**. If there were no link, a correlation this strong in 300 people would happen less than 1 time in 1000.

TRAP

p tells you whether an effect exists, not how big or important it is. With 300 people even small effects can be significant. For size, look at r, B, β or the odds ratio.

3. The 95% confidence interval: "how sure are we?"

Our numbers come from 300 people, not everyone. The 95% CI is the range where the true value probably lies. Narrow = precise.

TWO RULES YOU WILL USE ALL THE TIME

Odds ratio: if the CI includes **1** → not significant. Example: chronic condition AOR 1.51 (0.69–3.30) includes 1 → not significant.

Difference or coefficient (B): if the CI includes **0** → not significant. Example: Female B −0.05 (−1.82 to 1.72) includes 0 → not significant.

4. Effect size: "how big?"

Measure	Used for	Small / medium / large	Ours
r (correlation)	two scores	0.1 / 0.3 / 0.5	0.47: moderate, close to large
Cramér's V	χ^2 tests	0.1 / 0.3 / 0.5	0.12–0.26: small
R^2	regression	share explained, 0–1	0.317: 32% of differences in CSS explained
Odds ratio	yes/no outcomes	1 = none; further from 1 = bigger	3.12 per SD of cyberchondria

5. Unadjusted versus adjusted

Unadjusted (bivariate, Tables 5–7): one variable at a time. **Adjusted** (multivariable, Tables 8–10): many variables in one model, so each result is "holding the others constant". A factor that is significant alone can disappear after adjustment if another factor explains it, and the reverse can happen too (you will see both in Table 8).

6. Normal or not normal

Many tests (t-test, ANOVA) assume scores form a bell curve. The **Shapiro–Wilk test** checks this. Ours were **not** normal (SHAI $p < 0.001$; CSS $p = 0.008$), so groups were compared with **non-parametric** tests (Mann–Whitney, Kruskal–Wallis), which compare ranks instead of means. Correlation and regression were kept because $n = 300$ is large, Spearman gave the same answer as Pearson, and the regression residuals were close to normal (checked with a histogram and P-P plot).

> **IF THEY ASK "WHY NO ANOVA OR T-TEST?"**
>
> "Shapiro–Wilk showed our scores were not normally distributed and some groups were small, so we used Kruskal–Wallis, the non-parametric version of one-way ANOVA, and Mann–Whitney instead of the t-test. We checked: ANOVA and t-tests give the same conclusions."

Part B · Which test, and why

The whole chapter uses nine tests. The choice follows one rule: **what kind of variable is compared with what**.

You want to know...	Variable types	Test	Where
How common is something, and how precise is that?	one categorical	**% with 95% CI (Wilson)**	Tables 1, 2, 4
Do the items of a scale measure one thing?	items of one scale	**Cronbach's α**	Table 3
Are the scores normal?	one score	**Shapiro–Wilk**	Methods
Do two scores rise together?	score × score	**Pearson r (Spearman ρ to confirm); partial r adjusted**	Table 5
Does a score differ between 2 groups?	score × 2 groups	**Mann–Whitney U**	Table 6
Does a score differ between 3+ groups?	score × 3+ groups	**Kruskal–Wallis H**	Table 6
Is a yes/no outcome related to a group?	categorical × categorical	**χ^2 (Fisher if cells small), Cramér's V**	Table 7
Which factors predict a score, together?	score ← many	**Hierarchical multiple linear regression**	Table 8
Which factors predict a yes/no, together?	yes/no ← many	**Binary logistic regression**	Tables 9–10
Does part of a link pass through a middle step?	score → score → score	**Mediation (Baron & Kenny) + Sobel test**	Table 11

THE LOGIC OF THE CHAPTER

The chapter moves from simple to complex: **describe** (3.1–3.2) → **one link at a time** (3.3–3.4) → **all factors together** (3.5) → **behaviour and pathways** (3.6). Each step answers a question the previous one leaves open.

SAY IT

All analyses were done in SPSS 26 with two-sided $p < 0.05$. Because our scores were not normally distributed we compared groups with Mann–Whitney and Kruskal–Wallis, used χ^2 for categories, Pearson and Spearman for the main correlation, and multiple linear and binary logistic regression for independent effects.

3.1 Participants and online searching

QUESTION	TEST	WHY THIS TEST
Who answered, how do they search, and what do they do afterwards?	Descriptive statistics only: n and %	Every study describes its sample first so the reader knows who the results apply to. No test is needed: we are describing, not comparing.

Table 1 · Socio-demographic characteristics

Characteristic	n	%
Age group (years)		
18-19	56	18.7
20-29	187	62.3
30-39	21	7.0
40-49	25	8.3
50-59	9	3.0
60+	2	0.7
Gender		
Female	169	56.3
Male	131	43.7
Highest education		
Currently a student	189	63.0
High school or below	18	6.0
Diploma/Vocational	7	2.3
Bachelor	65	21.7
Postgraduate	21	7.0
Study/work		
Medical student	132	44.0
Other health-science student	43	14.3
Healthcare worker	15	5.0
Non-health student	46	15.3
Non-health employee/other	64	21.3
Health-related background (1=yes)		
Yes	190	63.3
No	110	36.7
Diagnosed chronic condition (1=yes)		
Yes	65	21.7
No	235	78.3
Language relied on		
Arabic	212	70.7
English	88	29.3

n	number of people in that row
%	n ÷ 300 × 100. Each block adds up to 300 people and 100%.

Walk-through

1. 20–29 years: 187 people → 187 ÷ 300 = **62.3%**. Add 18–19 (18.7%) and **81%** are under 30.

2. Medical students: 132 → **44.0%**. Health-related background (medical + other health students + health workers): 190 → **63.3%**.
3. Female 56.3%; chronic condition 21.7%; answered mostly in Arabic 70.7%.

SAY IT

Our 300 participants were young, 62% aged 20–29, and 44% were medical students, so our percentages describe a young, educated online sample, not all of Baghdad.

IF THEY ASK

Why so many medical students?

The form spread through our own networks first (convenience and snowball sampling). We state this as a limitation and do not generalise percentages; associations are still informative.

Table 2 · Digital health-information seeking

Variable	n	%
Tools used (multiple)		
AI chatbots*	266	88.7
Search engines	169	56.3
Dedicated medical platforms	55	18.3
Social media	32	10.7
Main AI chatbot		
ChatGPT	178	59.3
Gemini	71	23.7
Claude	12	4.0
Other AI	3	1.0
None reported	36	12.0
Tool profile		
AI + search engine	143	47.7
AI without search engine	123	41.0
No AI (search/other only)	34	11.3
Motivation (multiple)		
Convenience/accessibility	240	80.0
Check if doctor visit is needed	173	57.7
Instant reassurance	153	51.0
Self-diagnose / double-check prescription	69	23.0
Cost/logistic barriers	54	18.0
Trust in online/AI medical data (1-5)		
1 – Do not trust at all	20	6.7
2 – Slightly	83	27.7
3 – Moderately	170	56.7
4 – Very much	25	8.3
5 – Completely	2	0.7
Behaviour after searching (multiple)		
Delay/skip care (reassured)	104	34.7
Ignore symptom	90	30.0
Visit multiple doctors	76	25.3
Urgent visit/ER/fast diagnosis	66	22.0
Self-medication	64	21.3
Delay care (terrified/overwhelmed)	26	8.7
Disclose search to physician		
Always	81	27.0
Sometimes	124	41.3
Never (fear of judgement)	95	31.7

Effect on doctor relationship		
No negative impact	196	65.3
Somewhat complicated	66	22.0
Significantly reduced trust	38	12.7

“multiple”	people could tick several options, so these blocks add to more than 100%
AI chatbots*	88.7% = ticked AI *or* named a chatbot. 78.0% ticked the AI box itself (the asterisk).
Tool profile	everyone placed in one group: AI + search engine / AI only / no AI. This block does add to 100%.

Walk-through

1. AI chatbots: 78.0% ticked (88.7% including the 32 who named one) vs search engines **56.3%** → AI is now first.
2. 41.0% used AI *without* any search engine. ChatGPT 59.3%, Gemini 23.7%.
3. Motives: convenience 80.0%, “do I need a doctor?” 57.7%, instant reassurance 51.0%.
4. After searching: delayed or skipped care because reassured **34.7%**; ignored the symptom 30.0%; several doctors 25.3%; urgent visit 22.0%.
5. Told the doctor: always 27.0%, sometimes 41.3%, **never 31.7%** (fear of judgement).

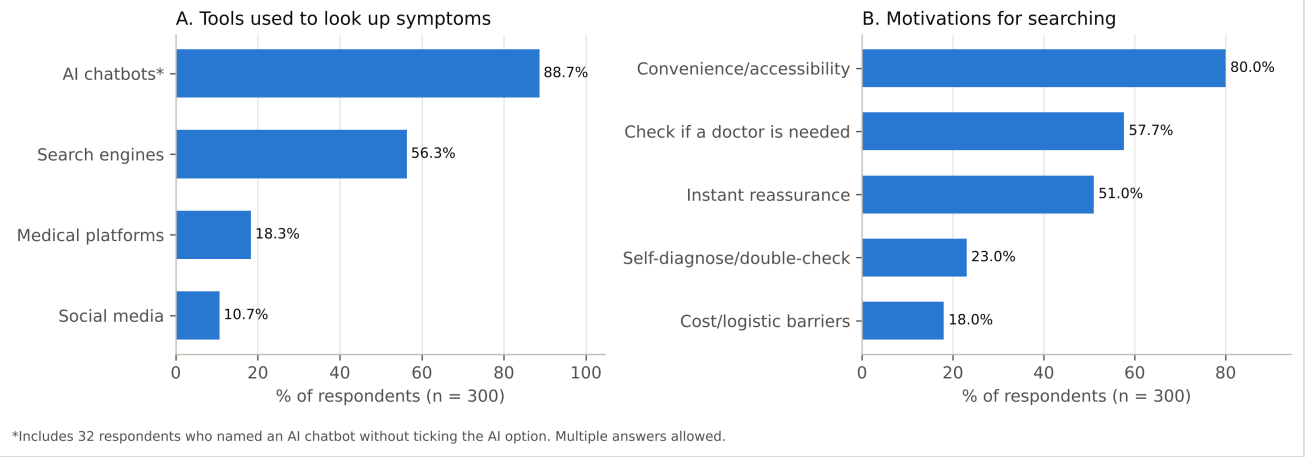


Figure 2: the same tools and motives as bars (each bar = % of all 300).

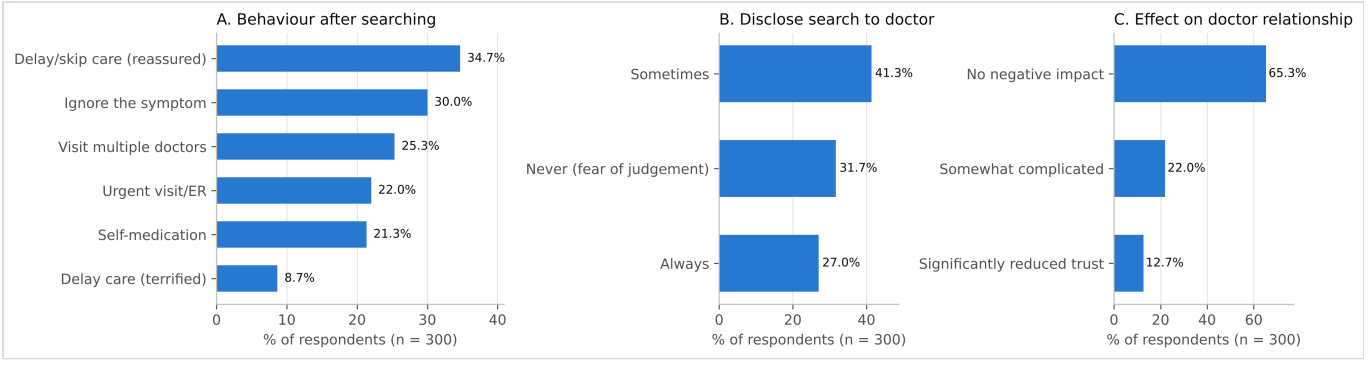


Figure 3: behaviour after searching, disclosure, and effect on the relationship with doctors.

SAY IT

AI chatbots are now the main gateway: 78% ticked them against 56% for search engines. After searching, one in three delayed care because they felt reassured, and almost one in three never tells the doctor.

TRAP

Two AI numbers exist: 78.0% (ticked the box) and 88.7% (ticked or named a chatbot). Always say which one. The difference is the 32 people who named a chatbot in the next question without ticking AI.

3.2 Levels of health anxiety and cyberchondria

QUESTION	TEST	WHY THIS TEST
How anxious are people, how much do they search, and do the scales work?	Mean ± SD, median (IQR), Cronbach's α; % with Wilson 95% CI	Before using any score you must show it is reliable (α) and describe its level. Prevalence needs a CI because it is an estimate.

Table 3 · Scores and reliability

Scale	Mean ± SD	Median (IQR)	Range	Cronbach's α
SHAI-14 total (0–42)	12.35 ± 5.83	12 (8–16)	1–30	0.81
CSS-12 total (12–60)	28.86 ± 8.36	29 (23–34)	12–57	0.86
Excessiveness (3–15)	9.66 ± 2.94	10 (8–12)	3–15	0.77
Compulsion (3–15)	5.00 ± 2.58	4 (3–6)	3–15	0.88
Distress (3–15)	7.24 ± 2.74	7 (5–9)	3–15	0.76
Reassurance (3–15)	6.96 ± 2.81	7 (5–9)	3–15	0.76

Mean ± SD	average and spread. About two-thirds of people lie within 1 SD of the mean.
Median (IQR)	the middle person, and the range of the middle half. Useful because scores are skewed.
Range	lowest and highest score anyone actually got (the possible range is in brackets)
Cronbach's α	do the items of the scale measure the same thing? ≥ 0.7 acceptable, ≥ 0.8 good, ≥ 0.9 excellent.

Walk-through

1. SHAI: mean **12.35 ± 5.83** out of 42 → most people scored about 6.5 to 18. Median 12 (half scored 8–16). Nobody scored above 30.
2. CSS: mean **28.86 ± 8.36** out of 12–60 → moderate searching.
3. Dimensions (each 3–15): excessiveness highest (9.66), compulsion lowest (5.00, median 4: most people are near the minimum of 3) → people search a lot, but it rarely disrupts work or social life.
4. α = 0.81 (SHAI) and 0.86 (CSS): both **good**. All four dimensions ≥ 0.76.

Table 4 · Prevalence at the published cut-offs

SHAI-14 score	Basis of the cut-off	n	%	95% CI
≥ 18	Optimal versus healthy controls (primary definition)	57	19.0	15.0–23.8
≥ 22	Optimal in a psychiatric setting	22	7.3	4.9–10.9
≥ 29	Optimal in a low-prevalence setting	2	0.7	0.2–2.4

Cut-off	the three thresholds from the Swedish validation study (Österman 2022). We use ≥ 18.
95% CI	Wilson interval: the true prevalence is probably between 15.0% and 23.8%.

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57 of 300 scored ≥ 18  →  57 ÷ 300 = 19.0%
95% CI 15.0% to 23.8%  (Wilson score interval)
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SAY IT

Both questionnaires were reliable, alpha 0.81 and 0.86. About one in five, 19%, scored in the elevated range of health anxiety, a screening result, not a diagnosis.

TRAP

Never say “19% have health anxiety disorder”. The SHAI is a screening questionnaire; a diagnosis needs a clinical interview.

IF THEY ASK

Why 18?

It best separated people with health anxiety from healthy controls in the Swedish diagnostic study. We also show 22 and 29 so the reader sees how the estimate depends on the threshold.

Is α 0.81 good enough?

Yes: 0.7 is the usual minimum and 0.8 is good.

3.3 The main association

QUESTION	TEST	WHY THIS TEST
Do people with more health anxiety have more cyberchondria (H1), and does it hold after adjustment (H2)?	Pearson r with 95% CI; Spearman ρ; partial r; adjusted β	Both are continuous scores, so a correlation. Spearman checks the result does not depend on normality; partial r and β check it is not explained by other factors.

Table 5 · Correlations

Cyberchondria measure	Pearson r (95% CI)	Spearman ρ	p	Adjusted β
CSS-12 total	0.47 (0.38–0.55)	0.47	<0.001	0.48
Excessiveness	0.46 (0.37–0.55)	0.45	<0.001	0.44
Compulsion	0.25 (0.14–0.36)	0.32	<0.001	0.29
Distress	0.45 (0.36–0.54)	0.44	<0.001	0.44
Reassurance	0.24 (0.13–0.35)	0.23	<0.001	0.25

Pearson r (95% CI)	strength of the straight-line link, −1 to +1. CI by Fisher's z.
Spearman ρ	the same idea using ranks; a check that does not need normal data
p	all < 0.001
Adjusted β	the standardised effect of SHAI on that measure from a regression with all other factors (age, gender, occupation, chronic disease, language, tools, trust, motives)

Walk-through

1. CSS total: r = **0.47** (0.38–0.55). The whole CI is well above 0 → a real, moderate-to-large link.
2. $r^2 = 0.47 \times 0.47 =$ **0.22** → health anxiety accounts for about 22% of the differences in cyberchondria.
3. Spearman ρ = 0.47 → identical: the result does not depend on the method.
4. Partial r = 0.48 after removing age, gender, health background, chronic disease, trust and AI use → unchanged, so those are not the explanation (H2).
5. By dimension: excessiveness 0.46 and distress 0.45 are strongest; compulsion 0.25 and reassurance 0.24 weakest (as H1 predicted).

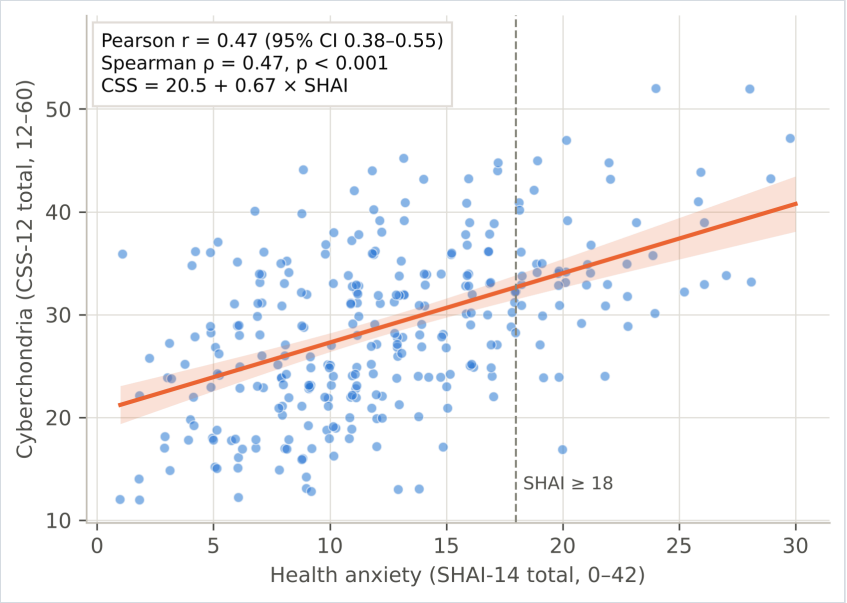


Figure 4: each dot is one participant; the orange line is the best fit (CSS = 20.5 + 0.67 × SHAI), the band its 95% CI.

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Reading the line: CSS = 20.5 + 0.67 × SHAI
SHAI 10  → 20.5 + 6.7  = 27.2
SHAI 20  → 20.5 + 13.4 = 33.9   (about 7 CSS points higher for 10 SHAI points)
```

SAY IT

This is our main result. Health anxiety and cyberchondria rise together, r = 0.47, and the link is the same after adjusting for everything else, 0.48. It is strongest for excessive searching and for distress after searching. Hypotheses 1 and 2 are supported.

TRAP

Say “associated with” or “linked to”. Never “health anxiety causes cyberchondria”: one survey at one time cannot show direction.

IF THEY ASK

Why is your r (0.47) lower than the meta-analyses (0.62)?

Our health anxiety scores were low and narrowly spread, our participants were confident internet users, and we used the short 12-item scale; all of these weaken correlations.

Pearson or Spearman, which is right?

Both: Pearson is the standard for two scores, Spearman checks it without assuming normality, and they agree (0.47).

3.4 Which groups differ

QUESTION	TEST	WHY THIS TEST
Are scores higher in some groups (age, chronic illness, AI users…)?	Mann–Whitney U (2 groups), Kruskal–Wallis H (3+ groups)	A score compared across groups, and the scores are not normal → the non-parametric versions of the t-test and one-way ANOVA.

Table 6 · Scores by participant characteristics (selected rows)

Variable	n	SHAI mean ± SD	p	CSS mean ± SD	p
Age group			0.008		0.288
18-19	56	14.0 ± 5.5		28.6 ± 8.1	
20-29	187	12.4 ± 5.8		28.4 ± 8.0	
30-39	21	10.2 ± 6.4		32.5 ± 11.3	
≥40	36	11.0 ± 5.7		29.7 ± 8.2	
Gender			0.378		0.859
Female	169	12.7 ± 6.0		29.0 ± 8.7	
Male	131	11.9 ± 5.5		28.6 ± 8.0	
Education			0.002		0.779
Currently a student	189	13.3 ± 5.8		28.6 ± 8.1	
High school or below	18	12.1 ± 5.4		26.9 ± 6.6	
Diploma/Vocational	7	9.3 ± 2.7		29.1 ± 8.3	
Bachelor	65	10.9 ± 5.9		29.7 ± 8.8	
Postgraduate	21	9.7 ± 5.3		30.0 ± 10.7	
Study/work			0.005		0.820
Medical student	132	13.3 ± 5.8		28.9 ± 8.1	
Other health-science student	43	12.2 ± 5.5		29.2 ± 9.1	
Healthcare worker	15	10.4 ± 5.1		29.5 ± 7.4	
Non-health student	46	13.0 ± 6.0		27.6 ± 7.8	
Non-health employee/other	64	10.5 ± 5.7		29.4 ± 9.1	
Chronic condition			0.004		0.435
Yes	65	14.2 ± 6.0		29.9 ± 8.9	
No	235	11.9 ± 5.7		28.6 ± 8.2	
Uses AI chatbots			0.008		<0.001
Yes	266	12.6 ± 5.7		29.5 ± 8.2	
No	34	10.5 ± 6.7		24.0 ± 8.1	
Uses social media			0.001		0.021
Yes	32	15.8 ± 6.9		32.5 ± 9.1	
No	268	11.9 ± 5.6		28.4 ± 8.2	
Tool profile			0.006		<0.001
AI + search engine	143	13.1 ± 5.6		30.1 ± 7.6	
AI without search engine	123	12.0 ± 5.8		28.7 ± 8.8	
No AI (search/other only)	34	10.5 ± 6.7		24.0 ± 8.1	
Motive: reassurance			<0.001		0.085
Yes	153	13.8 ± 5.7		29.6 ± 7.6	
No	147	10.9 ± 5.7		28.1 ± 9.1	

n	people in that group
SHAI / CSS mean ± SD	average score of that group
p (on the bold variable row)	one p for the whole variable: Mann–Whitney if 2 groups, Kruskal–Wallis if 3 or more. Below 0.05 = the groups differ.

Walk-through

1. **Chronic condition**: SHAI 14.2 vs 11.9, Mann–Whitney **p = 0.004** → people with a chronic illness are more health-anxious. CSS 29.9 vs 28.6, p = 0.435 → but they do not search more.
2. **Reassurance motive**: SHAI 13.8 vs 10.9, **p < 0.001**, the clearest difference in the table.
3. **AI users**: CSS 29.5 vs 24.0, **p < 0.001**. **Tool profile**: 30.1 (AI + search) vs 28.7 (AI only) vs 24.0 (no AI), Kruskal–Wallis p < 0.001 (Figure 5).
4. **Age**: SHAI falls from 14.0 at 18–19 to 10.2 at 30–39, p = 0.008 → younger people are more anxious. Education and occupation also differ for SHAI (students highest).
5. **Gender**: p = 0.378 and 0.859 → no difference.

> **TRAP**
> Kruskal–Wallis tells you that the groups differ somewhere, not which pair. Read the means to see the pattern (e.g. no-AI users clearly lowest). The thesis does not claim specific pairs.

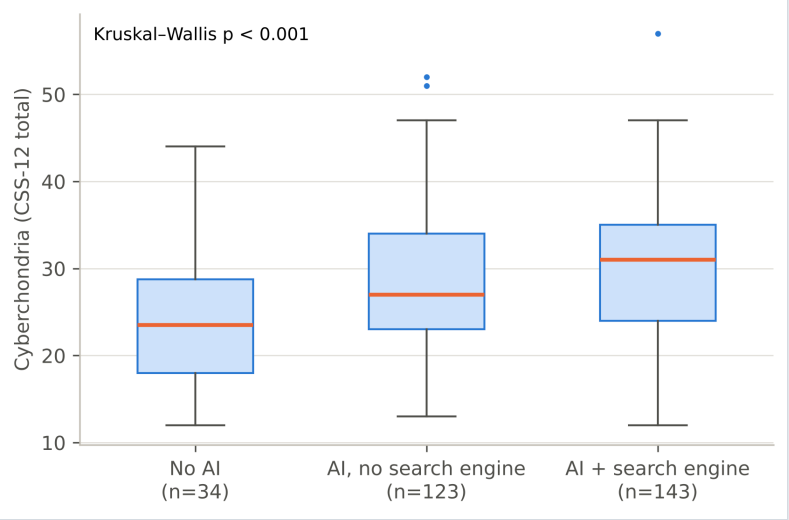


Figure 5: box plot of CSS by tool profile. Line = median; box = middle half of people; whiskers = range.

Table 7 · Yes/no outcomes (χ^2), significant rows only

Variable	n/N (%) with outcome, by group	Cramér's V	p
Elevated health anxiety (SHAI ≥ 18)			
Motive: reassurance	No: 15/147 (10.2%); Yes: 42/153 (27.5%)	0.22	<0.001
Uses social media	No: 44/268 (16.4%); Yes: 13/32 (40.6%)	0.19	0.001
Delay care (terrified)	No: 47/274 (17.2%); Yes: 10/26 (38.5%)	0.15	0.016
Motive: convenience	No: 17/60 (28.3%); Yes: 40/240 (16.7%)	0.12	0.039
High cyberchondria (CSS > 33)			
Elevated health anxiety	No: 56/243 (23.0%); Yes: 30/57 (52.6%)	0.26	<0.001
Disclose to physician	Always: 33/81 (40.7%); Never : 15/95 (15.8%); Sometimes: 38/124 (30.6%)	0.21	0.001
Ignore symptom	No: 70/210 (33.3%); Yes: 16/90 (17.8%)	0.16	0.006
Uses AI chatbots	No: 3/34 (8.8%); Yes: 83/266 (31.2%)	0.16	0.007
Tool profile	AI + search engine: 49/143 (34.3%); AI without search engine: 34/123 (27.6%); No AI (search/other only): 3/34 (8.8%)	0.17	0.012
Effect on doctor relationship	No negative impact: 46/196 (23.5%); Significantly reduced trust: 14/38 (36.8%); Somewhat complicated: 26/66 (39.4%)	0.16	0.023
Main AI chatbot	ChatGPT: 61/178 (34.3%); Claude: 4/12 (33.3%); Gemini: 17/71 (23.9%); None reported: 4/36 (11.1%); Other AI: 0/3 (0.0%)	0.18	0.034
Delay care (terrified)	No: 74/274 (27.0%); Yes: 12/26 (46.2%)	0.12	0.039

n/N (%) with outcome	in each group: how many had the outcome out of how many in the group
Cramér's V	strength of the association: 0.1 small, 0.3 medium
p	Pearson χ^2 or Fisher's exact test

Walk-through: the key row

High cyberchondria (CSS > 33) by elevated health anxiety:

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Elevated HA:     30 of 57  = 52.6% high CSS  → odds = 30/27  = 1.11
Not elevated:    56 of 243 = 23.0% high CSS  → odds = 56/187 = 0.30
Odds ratio = 1.11 / 0.30 = 3.71   (95% CI 2.04–6.76),  χ² p < 0.001,  V = 0.26
```

Also: elevated health anxiety was more common in reassurance seekers (27.5% vs 10.2%) and social-media users (40.6% vs 16.4%); high cyberchondria was more common in people who always tell their doctor (40.7%) than in those who never do (15.8%).

SAY IT

Health anxiety was higher in people with a chronic illness and, most clearly, in those searching for instant reassurance. AI users had higher cyberchondria, and people with elevated health anxiety were more than twice as likely to have high cyberchondria, 53% against 23%.

IF THEY ASK

Why is high cyberchondria “> 33”?

The CSS-12 has no validated cut-off, so we used the upper third of our sample.

Why Fisher sometimes?

χ^2 is unreliable when expected counts in a cell are below 5; Fisher's exact test is used then.

3.5 Independent predictors

QUESTION	TEST	WHY THIS TEST
Which factors predict cyberchondria when all are considered together (H2)?	Hierarchical multiple linear regression (enter method, 3 blocks)	The outcome (CSS) is a score and there are many predictors. Entering them in blocks shows how much each group of factors adds.

Table 8 · Multiple linear regression of CSS (final model)

Predictor	B	95% CI	β	p
Constant	12.04	6.66–17.42		<0.001
Age 18–19 (ref. 20–29)	−0.43	−2.85–2.00	−0.02	0.730
Age 30–39	6.27	2.42–10.11	0.19	0.001
Age ≥40	3.98	0.16–7.80	0.15	0.041
Healthcare worker (ref. non-health employed)	1.06	−3.27–5.39	0.03	0.630
Medical student	0.41	−2.83–3.64	0.02	0.805
Non-health student	−0.73	−4.40–2.95	−0.03	0.697
Other health-science student	1.32	−2.31–4.95	0.06	0.474
Female	−0.05	−1.82–1.72	0.00	0.954
Chronic condition	−0.14	−2.20–1.92	−0.01	0.895
Answered mainly in English	0.51	−1.45–2.48	0.03	0.607
Uses AI chatbots	5.33	2.57–8.09	0.20	<0.001
Uses search engines	0.95	−0.80–2.69	0.06	0.287
Uses medical platforms	0.17	−2.14–2.48	0.01	0.886
Uses social media	1.46	−1.31–4.24	0.05	0.299
Trust in online/AI (per point)	0.72	−0.44–1.87	0.06	0.225
Motive: instant reassurance	−1.11	−2.91–0.70	−0.07	0.228
Motive: self-diagnosis	0.99	−1.10–3.09	0.05	0.352
Motive: cost/logistics	0.44	−1.79–2.67	0.02	0.697
Motive: need for a doctor?	0.09	−1.64–1.83	0.01	0.916
Health anxiety (SHAI, per point)	0.68	0.53–0.84	0.48	<0.001

Block entered	R^2 (share explained)	Added (ΔR^2)	p for the added block
1 · Socio-demographics	0.030	3.0%	0.53 (not significant)
2 · + Digital use	0.133	+10.3%	< 0.001
3 · + Health anxiety	**0.317**	**+18.4%**	< 0.001

Final model: F = 6.48, p < 0.001; adjusted R^2 0.268; all VIF low.

B	how many CSS points change for 1 unit of the predictor, with all others held constant. For yes/no predictors: difference between "yes" and the reference group.
95% CI	if it crosses 0 → not significant
β	standardised B: lets you compare predictors measured on different scales. Bigger = stronger.
Constant	predicted CSS when every predictor is 0. Not meaningful here; ignore it.
ref.	the comparison group for categories: age 20–29; non-health employed

Walk-through

1. **Health anxiety**: B = 0.68 (0.53–0.84), β = **0.48**, p < 0.001 → each extra SHAI point adds 0.68 CSS points; 10 points more anxiety ≈ 7 points more cyberchondria. The largest β by far.
2. **AI users**: B = 5.33 → 5.3 CSS points higher than non-users, p < 0.001. But with the strict definition (ticked AI only) B = 1.49, p = 0.21 → **not robust**.
3. **Age 30–39** (B 6.27, p = 0.001) and **≥ 40** (B 3.98, p = 0.041) higher than 20–29. Why, when Table 6 showed no age difference in CSS (p = 0.288)? Because older people have *lower* health anxiety but the *same* cyberchondria; once anxiety is held constant, their searching is relatively higher. This is exactly what adjustment does.
4. Gender, occupation, chronic condition, language, trust and motives: all CIs cross 0 → no independent effect.

SAY IT

Who you are explains only 3% of cyberchondria; digital habits add 10%, and health anxiety adds another 18%, to 32% in total. Health anxiety is by far the strongest predictor, beta 0.48. The AI effect disappears when only people who ticked AI are counted.

IF THEY ASK

What is the difference between B and β?

B is in real units (CSS points per SHAI point); β puts every predictor on the same scale so their strength can be compared.

Why hierarchical?

To show what health anxiety adds after demographics and digital use are already in the model: 18.4%, the biggest block.

Table 9 · Binary logistic regression of elevated health anxiety

Predictor	AOR	95% CI	p
Age group (per step)	0.61	0.39–0.93	0.023
Female	1.23	0.62–2.43	0.558
Health-related background	1.05	0.48–2.27	0.905
Chronic condition	1.51	0.69–3.30	0.305
Answered mainly in English	0.94	0.45–1.96	0.870
Uses AI chatbots	0.61	0.18–2.13	0.442
Uses search engines	1.43	0.69–3.00	0.338
Trust in online/AI (per point)	0.74	0.46–1.19	0.217
Motive: instant reassurance	3.83	1.80–8.18	<0.001
Motive: self-diagnosis	0.93	0.43–2.03	0.863
Cyberchondria (per 1 SD)	3.12	2.11–4.62	<0.001

AOR	adjusted odds ratio: how many times the odds of elevated health anxiety change, holding the others constant. 1 = no effect; > 1 higher; < 1 lower.
95% CI	if it includes 1 → not significant
per 1 SD	cyberchondria was entered as a z-score; 1 SD = 8.4 CSS points
per step	one age group older (e.g. 20–29 → 30–39)

Walk-through

1. Cyberchondria: AOR **3.12** (2.11–4.62) → every 8 extra CSS points triple the odds of elevated health anxiety.
2. Searching for instant reassurance: AOR **3.83** (1.80–8.18) → almost four times the odds. The CI is wide because only 57 people had elevated anxiety.
3. Age: AOR **0.61** per step → each older age group has 39% lower odds (1 − 0.61 = 0.39): **protective**.
4. Chronic condition 1.51 (0.69–3.30) includes 1 → not significant here, even though it was significant on the SHAI score in Table 6 (splitting a score into yes/no loses information).

How good is the model?

Check	Ours	Meaning
Omnibus χ^2	$\chi^2(11) = 64.9$, $p < 0.001$	the model predicts better than nothing
Nagelkerke R^2	0.31	roughly "31% explained" (a logistic version of R^2)
Hosmer–Lemeshow	$p = 0.50$	here we **want** $p > 0.05$: predicted and observed agree → good fit
AUC (ROC curve)	0.83	0.5 = coin toss, 0.8–0.9 good: the model separates the two groups well
Correctly classified	81%	see the trap below

TRAP

81% correct sounds impressive, but 243 of 300 (81%) did **not** have elevated anxiety, so a "model" that always guessed "no" would also be 81% correct. That is why the AUC (0.83) is the number to quote for how well the model works.

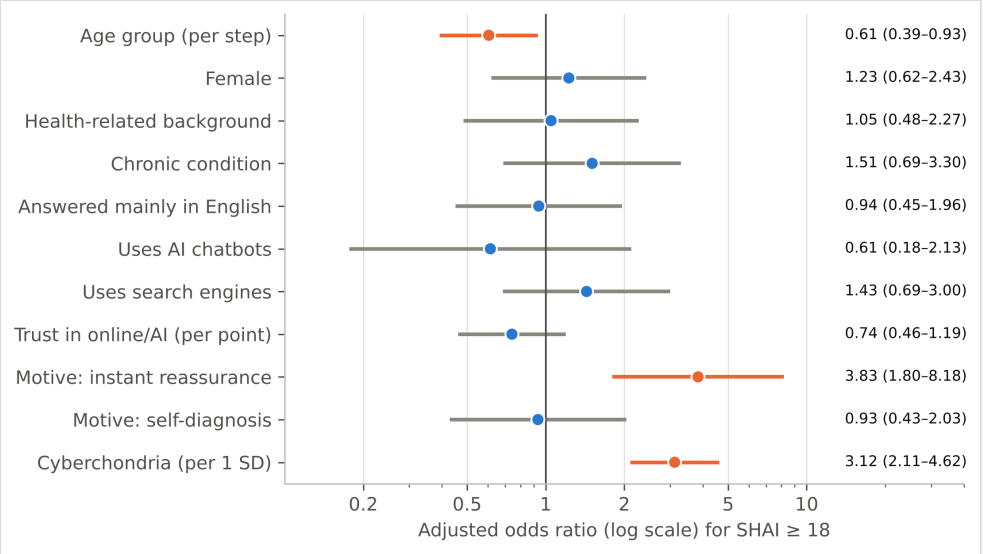


Figure 6: forest plot of Table 9. Dot = AOR, line = 95% CI, vertical line at 1 = no effect; orange = significant.

SAY IT

Three factors stand out: higher cyberchondria triples the odds of elevated health anxiety, searching for reassurance nearly quadruples them, and older age is protective. The model fits well and its AUC is 0.83.

IF THEY ASK

Why are the odds ratios so large and wide?

Only 57 people had elevated anxiety for 11 predictors, about 5 per variable, so the estimates are imprecise and may be overestimated. We say so in the limitations.

3.6 Behaviour and pathways

QUESTION	TEST	WHY THIS TEST
Which score goes with what people do before and during the consultation (H4)?	Binary logistic regression, one model per behaviour, both scores in the same model	Each behaviour is yes/no. Putting SHAI and CSS together shows which one matters once the other is held constant.

Table 10 · Behaviour, health anxiety and cyberchondria

Outcome	Events	SHAI AOR (95% CI)	p	CSS AOR (95% CI)	p
Urgent visit/ER	66	0.91 (0.66–1.25)	0.558	1.51 (1.09–2.09)	0.013
Multiple doctors	76	1.06 (0.78–1.44)	0.701	1.20 (0.88–1.65)	0.250
Delay/skip care (reassured)	104	0.89 (0.66–1.19)	0.423	0.83 (0.62–1.11)	0.211
Delay care (terrified)	26	1.49 (0.95–2.34)	0.082	1.59 (0.98–2.56)	0.059
Self-medication	64	0.81 (0.57–1.14)	0.220	1.00 (0.72–1.39)	0.994
Ignore symptom	90	1.16 (0.85–1.58)	0.357	0.65 (0.47–0.90)	0.009
Never disclose to physician	95	1.47 (1.07–2.00)	0.017	0.57 (0.41–0.80)	<0.001
Negative effect on doctor relationship	104	1.03 (0.78–1.37)	0.822	1.35 (1.01–1.81)	0.041

Events	how many said yes to that behaviour
SHAI AOR / CSS AOR	odds ratio per 1 SD of each score, each adjusted for the other and for trust, AI use, age, gender, health background, chronic disease and language

Walk-through

1. Urgent visit: CSS AOR **1.51**, p = 0.013 → 1 SD more cyberchondria = 51% higher odds of an urgent visit. SHAI 0.91, p = 0.558 → anxiety alone does not.
2. Ignore the symptom: CSS **0.65**, p = 0.009 → 35% lower odds: people with more cyberchondria do not ignore symptoms.
3. Never tell the doctor: CSS **0.57**, p < 0.001 → *more* disclosure; SHAI **1.47**, p = 0.017 → more *hiding*. The two scores pull in opposite directions.
4. Negative effect on the relationship: CSS **1.35**, p = 0.041.
5. Several doctors, self-medication and delay: not significant (fear-driven delay borderline, p = 0.059).
6. Trust in online/AI information (in the text): fewer multiple-doctor visits, AOR 0.59, p = 0.004.

```
Reading an AOR below 1:  AOR 0.57  →  1 − 0.57 = 0.43  →  43% lower odds of never disclosing
```

SAY IT

Behaviour goes with cyberchondria more than with health anxiety: higher cyberchondria goes with urgent visits, a more strained relationship and more disclosure, while health anxiety goes with hiding searches from the doctor.

TRAP

These p-values were not corrected for the many tests run. Say “associated with” and, for the weaker ones (0.01–0.05), “may”. Hypothesis 4 is only partly supported.

Table 11 · Mediation (pathways)

QUESTION	TEST	WHY THIS TEST
Does part of the link from health anxiety to distress pass through excessive searching (H3)?	Mediation by linear regressions (Baron & Kenny) + Sobel test	Mediation asks whether a middle variable carries part of a link. Sobel tests whether that indirect part is bigger than zero.

Model	a (SE)	b (SE)	Total c	Direct c′	Indirect a×b	Sobel z	p
HA → excessive searching → post-search distress	0.46 (0.05)	0.35 (0.06)	0.46	0.30	0.16	5.14	<0.001
HA → post-search distress → reassurance seeking	0.46 (0.05)	0.48 (0.06)	0.22	0.01 (ns)	0.22	5.98	<0.001
Excessive searching → post-search distress → HA (reverse order)	0.49 (0.05)	0.30 (0.06)	0.47	0.32	0.15	4.66	<0.001

a	predictor → mediator (health anxiety → excessive searching)
b	mediator → outcome, with the predictor held constant
Total c	the whole link, predictor → outcome
Direct c′	what is left of the link once the mediator is in the model
Indirect a×b	the part that passes through the mediator; equals c − c′
Sobel z, p	is the indirect part bigger than 0? |z| > 1.96 → p < 0.05
(SE)	standard error; all variables were z-scores, so coefficients are standardised

Walk-through: model A, step by step

1. Step 1 · a: health anxiety → excessive searching = **0.46** (significant).
2. Step 2 · b: excessive searching → distress, holding anxiety constant = **0.35** (significant).
3. Step 3 · total c = **0.46**; direct c′ = **0.30** → the link shrinks but does not disappear.
4. Indirect = a × b = 0.46 × 0.35 = **0.16** = c − c′. Share carried: 0.16 ÷ 0.46 ≈ **35%.**

```
Sobel z = (a × b) / √(b²·SEa² + a²·SEb²)
        = (0.462 × 0.352) / √(0.352² × 0.051² + 0.462² × 0.056²)
        = 5.14   → far above 1.96 → p < 0.001
```

Model B (anxiety → distress → reassurance seeking): total c = 0.22, direct c′ = 0.01 (not significant) → almost **all** of the link passes through distress (Sobel z = 5.98).

Reverse order (searching → distress → anxiety): also significant (indirect 0.15, z = 4.66). So the data cannot tell which comes first.

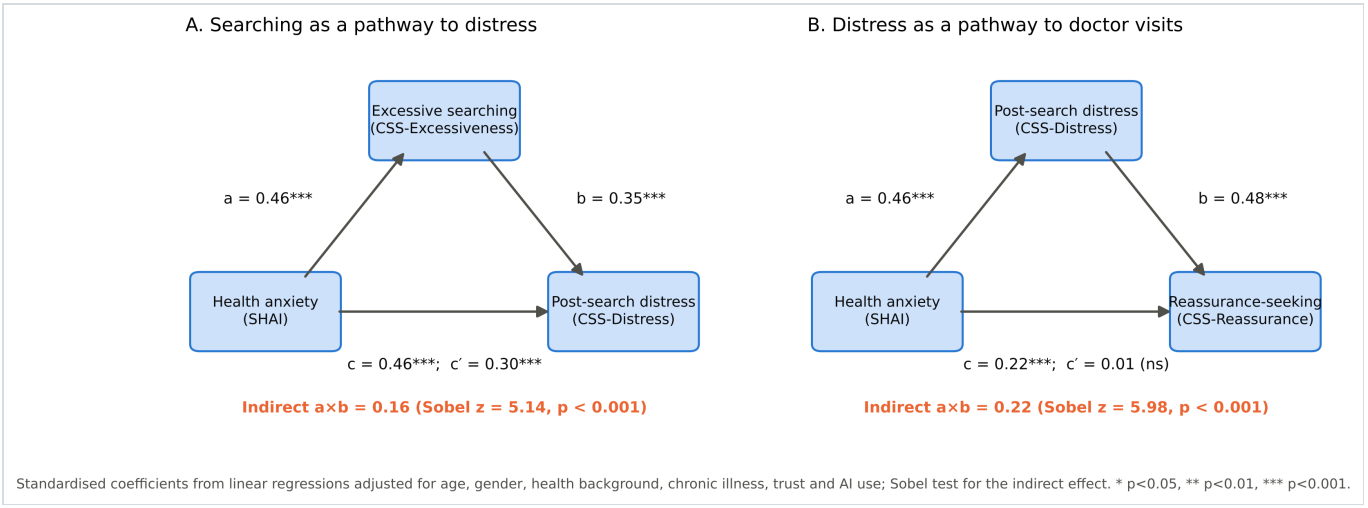


*Figure 7: the two mediation models drawn as path diagrams. *** p < 0.001; ns = not significant.*

SAY IT

About a third of the link between health anxiety and distress after searching passes through excessive searching, and almost all of the link to reassurance seeking passes through distress. This fits a vicious cycle, but because the reverse order also fits, one survey cannot show which comes first. Hypothesis 3 is supported statistically, not causally.

IF THEY ASK

Does this prove a vicious cycle?

No. It is one time point, the reverse model fits too, and the mediator and outcome come from the same questionnaire. It is compatible with a cycle; only longitudinal data can show direction.

Why Sobel and not bootstrap?

Sobel is the standard test that SPSS users report with the Baron and Kenny steps; all our indirect effects were far from zero (z 4.7–6.0).

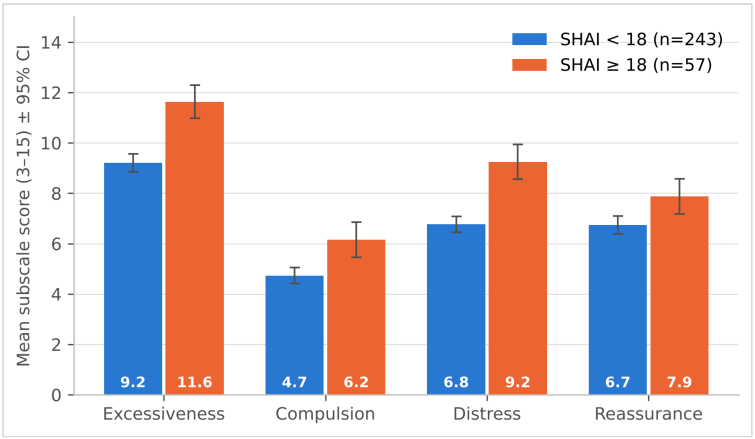


Figure 8: the four CSS dimensions by health-anxiety status (mean ± 95% CI). Biggest gaps: excessiveness 9.2 vs 11.6 and distress 6.8 vs 9.3.

Part C · The hypotheses, with the evidence

Hypothesis	Verdict	Evidence
H1 · More health anxiety → more cyberchondria, especially distress and excessiveness	**Supported**	r = 0.47 (0.38–0.55); excessiveness 0.46 and distress 0.45 strongest (Table 5, Figure 4)
H2 · Holds after adjustment	**Supported**	partial r = 0.48; health anxiety adds 18.4%, β = 0.48 (Table 8)
H3 · Excessive searching carries part of the link to distress	**Supported statistically, not causally**	indirect 0.16, Sobel z 5.14; reverse order also significant (Table 11, Figure 7)
H4 · Cyberchondria → urgent visits, several doctors, strained relationship	**Partly supported**	urgent visits (AOR 1.51) and strained relationship (1.35) yes; several doctors no (Table 10)

> **SAY IT**
> Hypotheses 1 and 2 were supported, hypothesis 3 statistically but not causally, and hypothesis 4 partly.

The never-say list

Never say	Say instead
Health anxiety causes cyberchondria	is associated with / is linked to
19% have health anxiety disorder	19% scored in the elevated range on a screening tool
AI makes people anxious	we found no reliable independent effect of AI use
We proved a vicious cycle	the results are compatible with a cycle
Our results represent Baghdad	a young online convenience sample
81% accuracy shows the model is good	the AUC of 0.83 shows good discrimination

Part D · Practice: read the row yourself

Cover the answers (next page). Say each answer out loud in one or two sentences.

1. Table 6: social-media users SHAI 15.8 vs 11.9, p = 0.001. Interpret.
2. Table 8: “Female B −0.05 (−1.82 to 1.72), p = 0.954.” Interpret.
3. Table 9: “Chronic condition AOR 1.51 (0.69–3.30).” Significant?
4. Table 10: SHAI and never disclosing: AOR 1.47, p = 0.017. Interpret.
5. What does $r^2 = 0.22$ mean?
6. A Kruskal–Wallis p = 0.008 for SHAI by age. Which age group differs?
7. Hosmer–Lemeshow p = 0.50. Good or bad?
8. Model B: c = 0.22, $c' = 0.01$. What does that tell you?
9. Table 7: V = 0.26. How strong?
10. Why is age significant in Table 8 but not for CSS in Table 6?

Answers

1. Social-media users have higher health anxiety; Mann–Whitney p = 0.001, so unlikely to be chance.
2. No difference between women and men once other factors are held constant; the CI includes 0.
3. No: the CI includes 1.
4. Each SD more health anxiety = 47% higher odds of never telling the doctor about searches (uncorrected, so “may”).
5. Health anxiety accounts for about 22% of the differences in cyberchondria.
6. The test shows only that the groups differ overall; the means show the youngest (18–19, 14.0) highest.
7. Good: we want p > 0.05; the model fits.
8. Almost the whole link from anxiety to reassurance seeking passes through distress.
9. Small to medium (0.1 small, 0.3 medium).
10. Adjustment: older people have lower anxiety but similar searching; holding anxiety constant reveals their relatively higher cyberchondria.

One-page summary of numbers

What	Number	Where
Participants	300; 62.3% aged 20–29; 44.0% medical students	Table 1
AI ticked / search engines	78.0% (88.7% incl. named) / 56.3%	Table 2
Delay when reassured / never tell	34.7% / 31.7%	Table 2, Fig 3
SHAI / CSS mean	12.35 ± 5.83 / 28.86 ± 8.36	Table 3
Cronbach's α	0.81 / 0.86	Table 3
Elevated health anxiety	19.0% (15.0–23.8); 7.3% at 22; 0.7% at 29	Table 4
Main correlation	r 0.47 (0.38–0.55); ρ 0.47; partial 0.48; r^2 22%	Table 5
High CSS by elevated HA	52.6% vs 23.0%; OR 3.71	Table 7
Regression	R^2 0.030 → 0.133 → 0.317; SHAI B 0.68, β 0.48	Table 8
AI effect	+5.33 inclusive; +1.49, p 0.21 strict	Table 8
Logistic	AOR 3.12 / 3.83 / 0.61; AUC 0.83; H–L p 0.50	Table 9
Behaviour (CSS)	urgent 1.51; ignore 0.65; never disclose 0.57; strained 1.35	Table 10
Mediation	A indirect 0.16, z 5.14; B c′ 0.01, z 5.98; reverse z 4.66	Table 11